# Smart Home Lighting Automation System

*Design of a Motion- and Light-Based Automated Lighting Control System*

CodeAlpha Robotics & Automation Internship — Task 3

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## 1. Introduction

Automating everyday tasks such as lighting control is one of the simplest and most practical entry points into home automation and embedded systems design. This report presents the proposed design of a smart home lighting automation system that turns a lamp on or off based on two conditions: whether a person is present in the room and whether the room is dark. The system uses a PIR motion sensor and an LDR light sensor as inputs to an Arduino Uno microcontroller, which evaluates both conditions and drives a lamp through a relay module. This document describes the system design, working logic, and expected behavior; it is presented as a proposed design rather than a built and tested prototype.

## 2. Objectives

- Design an automation system that switches a light ON only when motion is detected and the room is dark.
- Avoid unnecessary energy use by keeping the light OFF during daylight or when the room is unoccupied.
- Demonstrate how two simple sensor inputs can be combined with conditional logic on a low-cost microcontroller.
- Produce a clear system block diagram and control-logic flowchart suitable for review and future implementation.

## 3. Components Used

- Arduino Uno — the microcontroller board that reads both sensors, runs the control logic, and drives the relay.
- PIR (Passive Infrared) motion sensor — detects human presence by sensing changes in infrared radiation emitted by a moving body.
- LDR (Light Dependent Resistor) — detects ambient light level; its resistance changes with light intensity and is read through an analog input.
- Relay module — an electrically operated switch that allows the low-voltage Arduino output to safely switch the lamp's higher-current circuit.
- LED/lamp — the output device representing the room light being controlled.

## 4. System Architecture

The system follows a simple linear architecture. The PIR sensor and the LDR both act as inputs, feeding data to the Arduino Uno, which acts as the central decision-making unit. Based on the combined sensor readings, the Arduino sends a control signal to the relay module, which physically switches the lamp on or off. Figure 1 illustrates this arrangement.

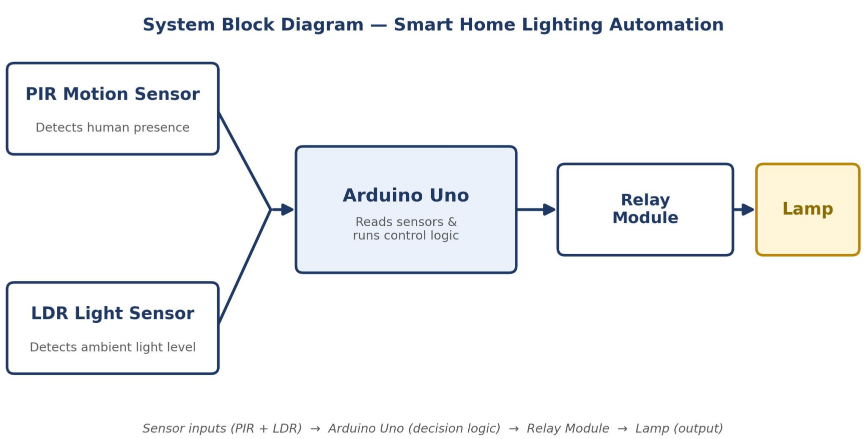


*Figure 1. System block diagram of the proposed smart lighting automation system.*

## 5. Working Principle

The PIR sensor continuously monitors the room for infrared changes caused by human movement; when a person moves within range, its output pin goes HIGH. At the same time, the LDR forms part of a voltage divider connected to an analog input pin, and its reading rises as ambient light falls. The Arduino reads both signals in its control loop, compares the LDR reading against a set brightness threshold, and checks the PIR output. Only when motion is present and the room is dark does the Arduino set its digital output HIGH, energizing the relay coil, which closes the switching contact and turns the lamp on. If either condition is not met, the output stays LOW and the lamp remains off.

## 6. Control Logic

The decision rule at the center of this design is a simple logical AND condition between the two sensor readings:

```
Light ON  =  (Motion Detected = YES)  AND  (Room Dark = YES)
```

If motion is absent, or the room is already bright, the lamp remains off regardless of the other condition. This prevents the two most wasteful scenarios in a simple motion-only system: lights turning on during the day, and lights staying on in a bright, occupied room where artificial lighting is not needed. The logic runs continuously in the Arduino's main loop, so the system re-evaluates both conditions every cycle rather than making a one-time decision.

## 7. Flowchart

Figure 2 shows the step-by-step control-logic flow implemented by the Arduino, from reading the two sensors to deciding the lamp state and looping back to monitor the room again.

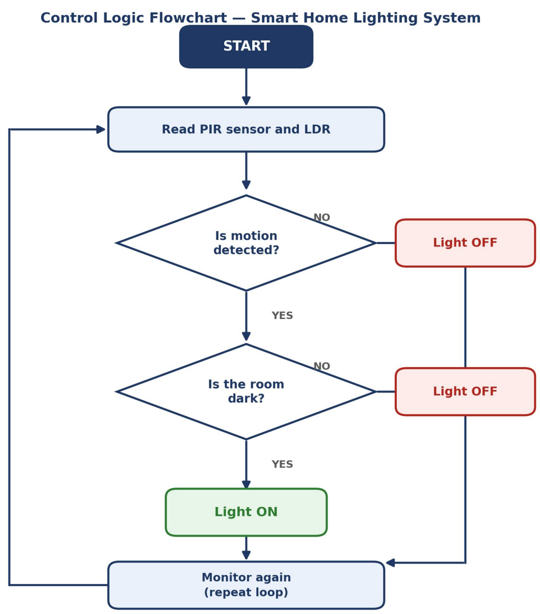


*Figure 2. Control logic flowchart for the smart lighting automation system.*

## 8. Advantages

- Reduces unnecessary energy consumption by keeping the light off during the day or when the room is empty.
- Requires no manual switching, which is convenient in spaces such as hallways, staircases, or storage rooms.
- Uses inexpensive, widely available components, making the design low-cost to prototype.
- The dual-condition logic is more reliable than a motion-only design, since it avoids false triggers in daylight.

## 9. Limitations

- A PIR sensor can miss very slow or minimal movement, which may cause the light to switch off while the room is still occupied.
- The LDR's brightness threshold may need manual calibration for different rooms and lighting conditions.

- The system reacts only to the two conditions programmed; it cannot distinguish between different occupants or account for personal preference.
- As with any relay-based design, the relay and wiring to mains-powered lamps must be handled according to proper electrical safety practice.

## 10. Future Improvements

- Add a short timer delay before switching off, so brief pauses in motion do not immediately turn off the light.
- Allow the brightness threshold to be adjusted through a potentiometer or a mobile app rather than fixed in code.
- Integrate the system with a Wi-Fi module (e.g., ESP8266) for remote monitoring and control.
- Extend the design to multiple rooms with a shared controller for whole-house automation.

## 11. Conclusion

This report has presented the design of a smart home lighting automation system that combines a PIR motion sensor and an LDR light sensor with an Arduino Uno and a relay module. By requiring both motion and darkness before switching a lamp on, the system avoids the common inefficiencies of simpler motion-only designs. The accompanying block diagram and flowchart lay out the system's structure and control logic clearly enough to guide a future physical build. As a proposed design, the next practical step would be prototyping and testing the circuit to confirm sensor thresholds and response times under real conditions.

## References

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